Indian Standard

53 GRADE ORDINARY PORTLAND CEMENT · SPECIFICATION

(Fifth Revision)

1 SCOPE

This standard covers the manufacturing, chemical and physical requirements of 53 grade ordinary portland cement.

2 REFERENCES

IS 4031 (Part 6) Methods of physical tests for hydraulic cement: Determination of compressive strength.
IS 4032 Method of chemical analysis of hydraulic cement.

3 TERMINOLOGY

For the purpose of this standard, the definitions given in IS 4845 shall apply.

4 MANUFACTURE

4.1 Cement shall be obtained by thoroughly grinding together clinker with calcium sulphate and suitable additions.

Chemical & Physical Requirements

5 CHEMICAL REQUIREMENTS

Clause 5.1 Chemical Composition Limits

When tested in accordance with IS 4032, 53 grade ordinary portland cement shall comply with the following:

a) Ratio of percentage of lime to percentages of silica, alumina and iron oxide (LSF): 0.80 to 1.02.

b) Ratio of percentage of alumina to that of iron oxide: Not less than 0.66.

c) Insoluble residue: Not exceeding 5.0 percent by mass.

d) Magnesia (MgO): Not exceeding 6.0 percent by mass.

e) Total sulphur content calculated as sulphuric anhydride (SO3): Not exceeding 3.5 percent by mass.

f) Total loss on ignition (LOI): Not exceeding 4.0 percent by mass.

6 PHYSICAL REQUIREMENTS

Clause 6.1 Fineness

Specific surface shall not be less than 225 m^2/kg when tested by Blaine air permeability method.

Clause 6.2 Compressive Strength Requirements

The average compressive strength of not less than three mortar cubes (area of face 50 cm^2) prepared in the manner described in IS 4031 (Part 6) shall be as follows:

a) 72 ± 1 h (3 days): Not less than 27 MPa (N/mm^2);

b) 168 ± 2 h (7 days): Not less than 37 MPa (N/mm^2);

c) 672 ± 4 h (28 days): Not less than 53 MPa (N/mm^2).

The cement cubes shall be tested at standard laboratory temperature of 27 ± 2°C.